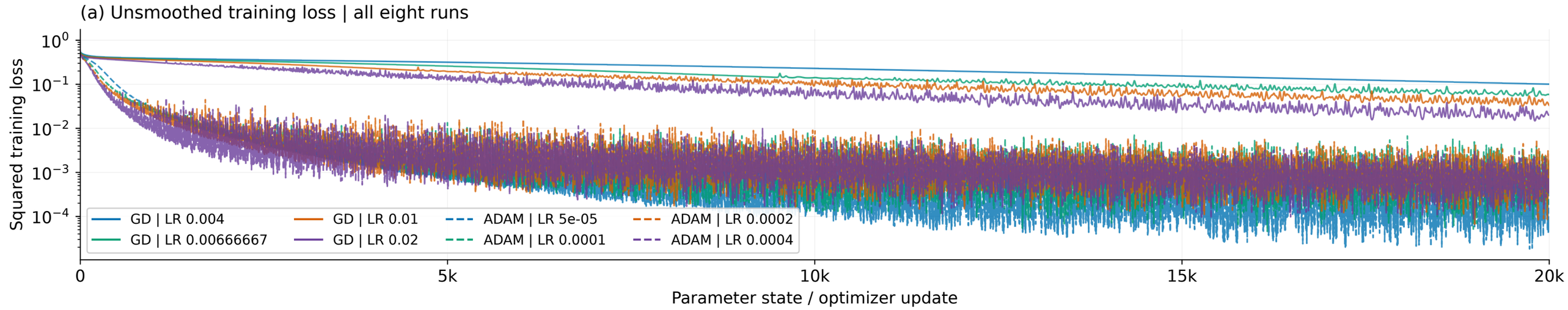
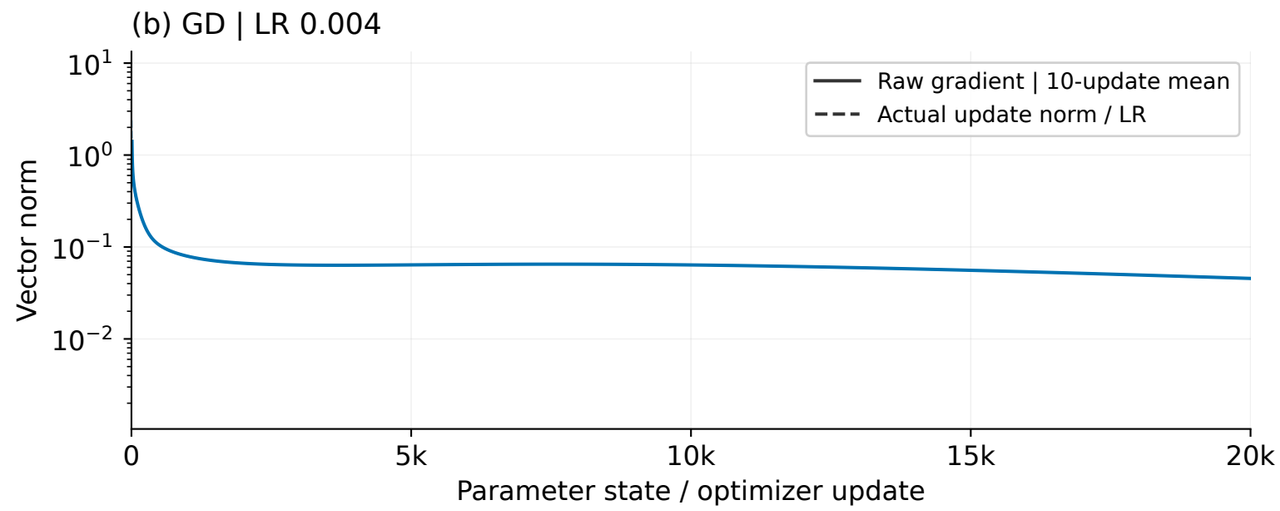
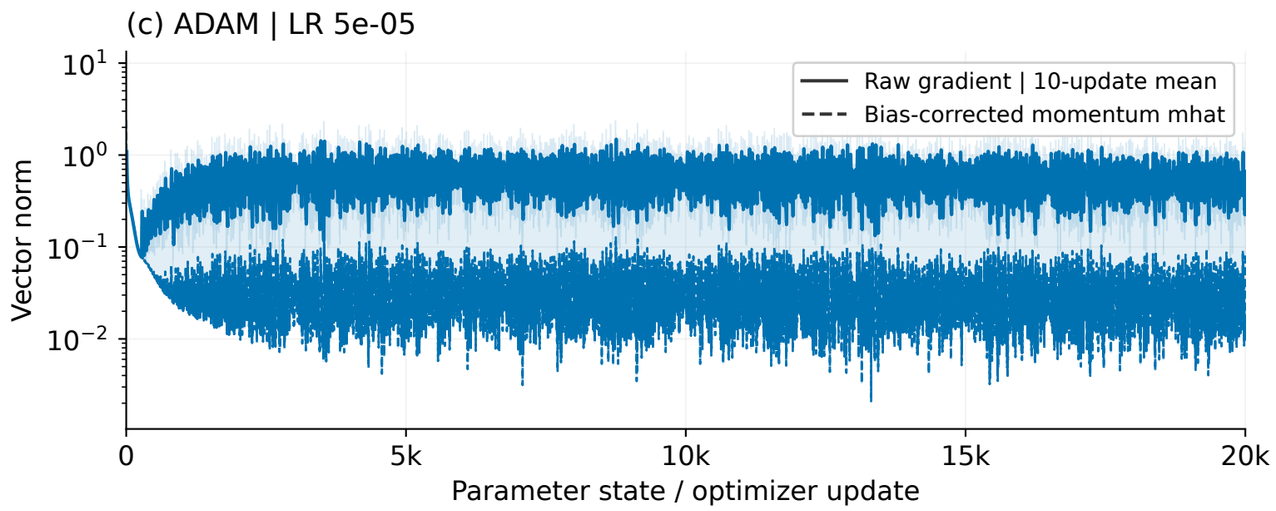
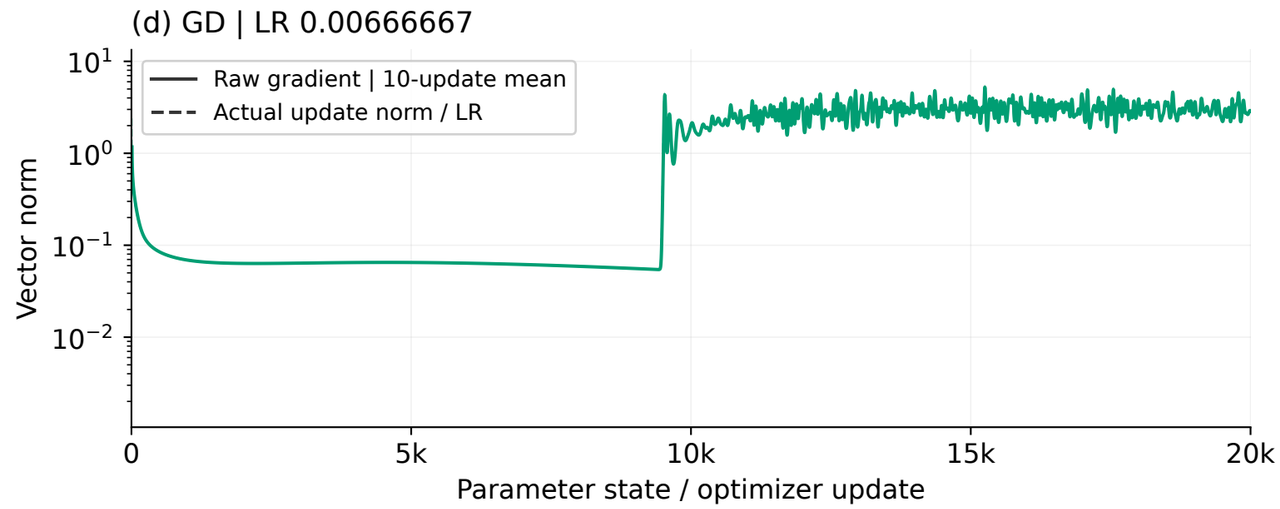
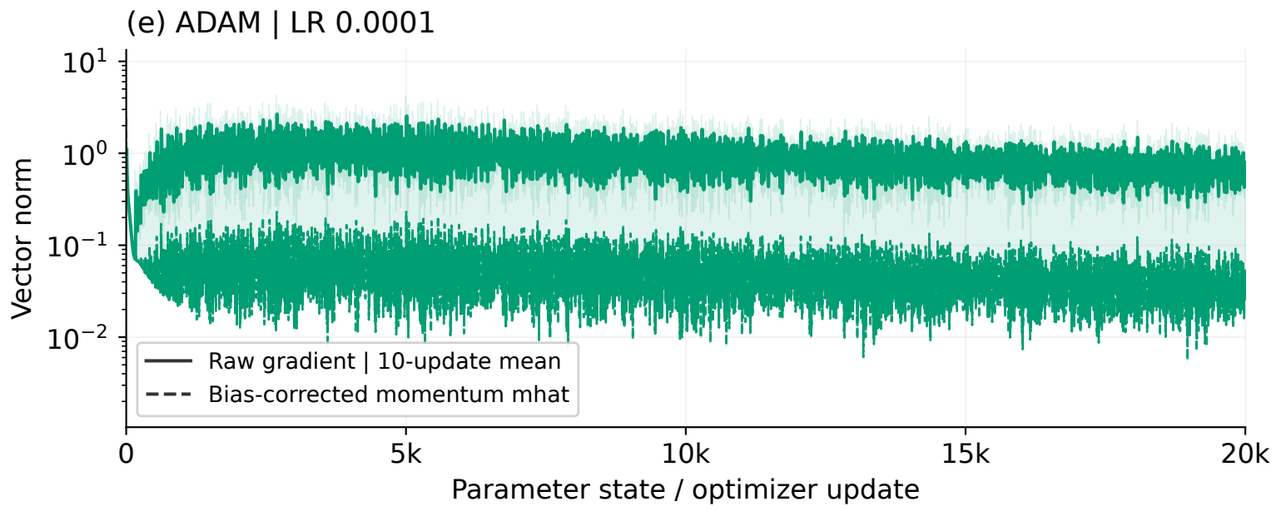
CIFAR-10 | loss and vector norms by learning rate
GD (left) and Adam (right) | common logarithmic norm scale

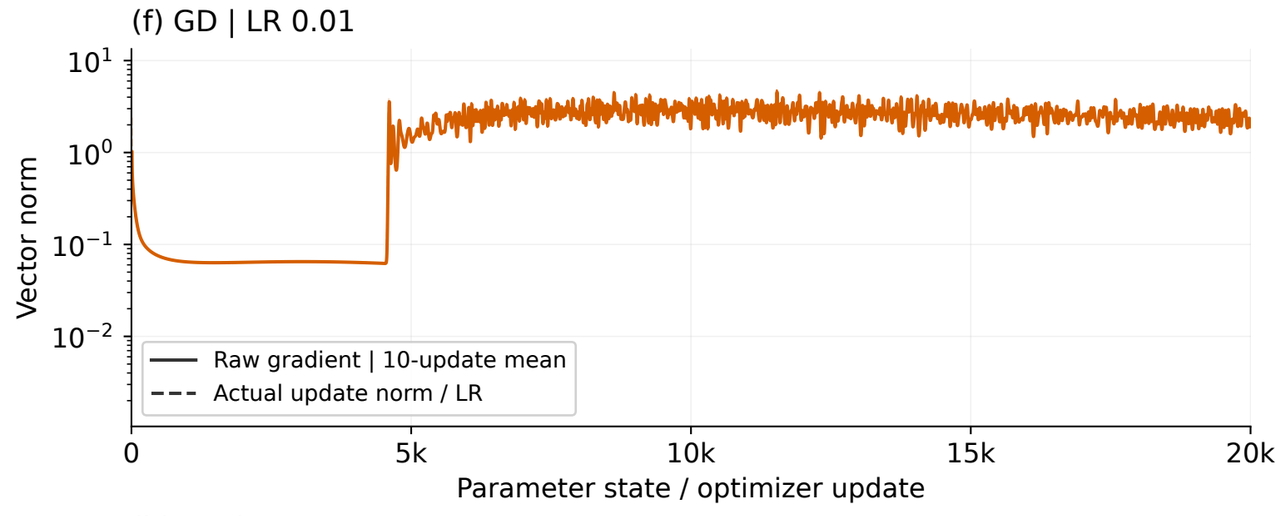


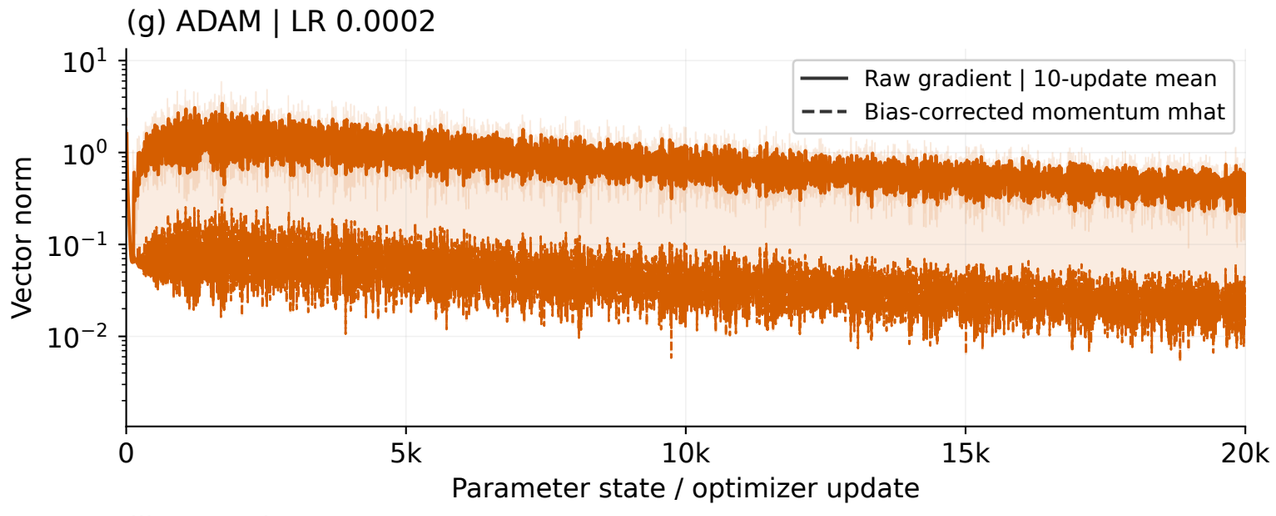


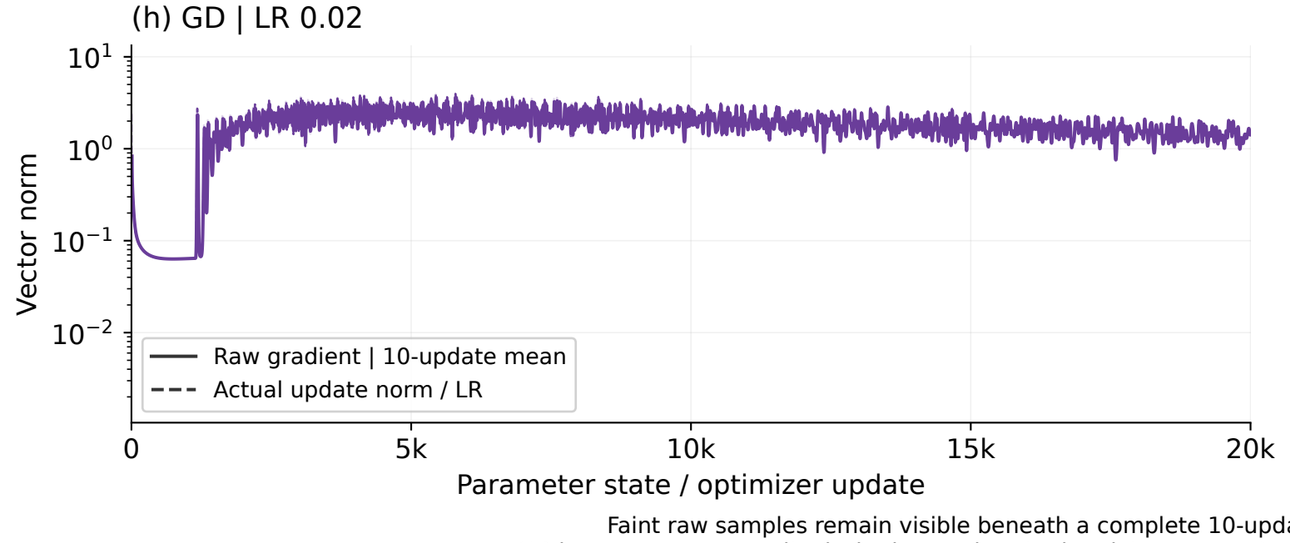


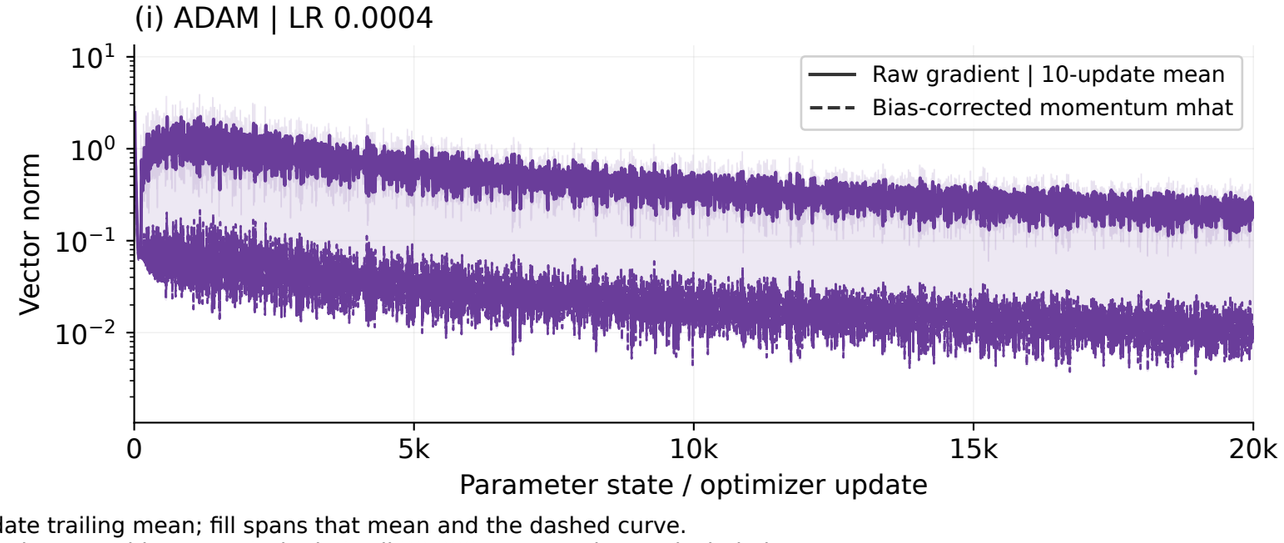












Faint raw samples remain visible beneath a complete 10-update trailing mean; fill spans that mean and the dashed curve.
GD has no momentum: its dashed curve is actual update norm / LR. Adam uses bias-corrected mhat. All 20,000 raw samples are included.